ISEL
DEETC
Departamento de Engenharia Electrónica e de Telecomunicações e de Computadores

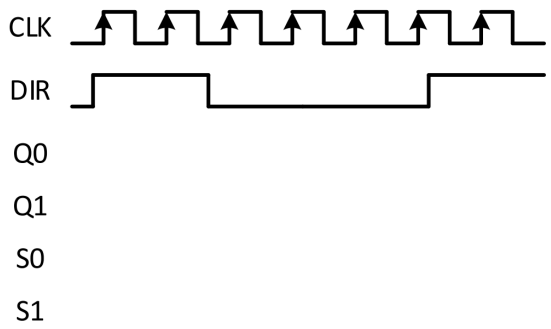
Aula 14

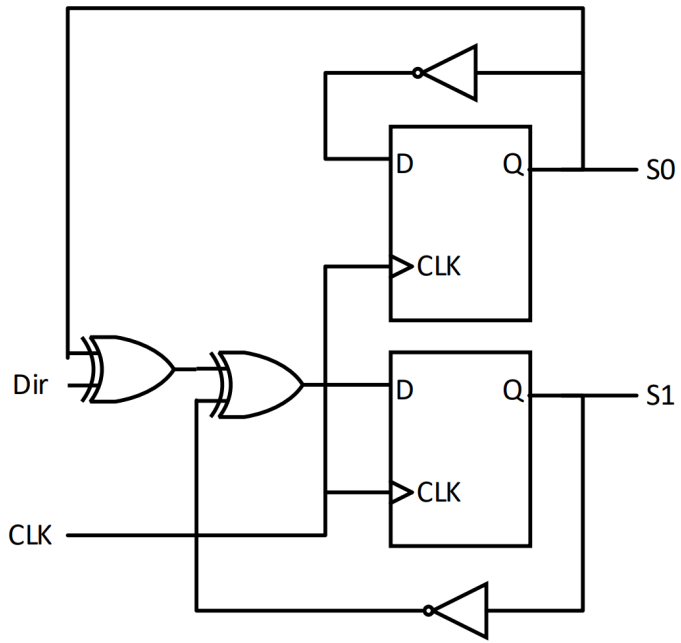
Sumário

- Circuitos sequenciais síncronos - Exercícios

Exercício 1

- Tendo em conta o circuito apresentado:
  - Determine, a tabela de transição de estados e de saídas
  - Determine, o ASM chart da máquina de estados
  - Complete o diagrama temporal

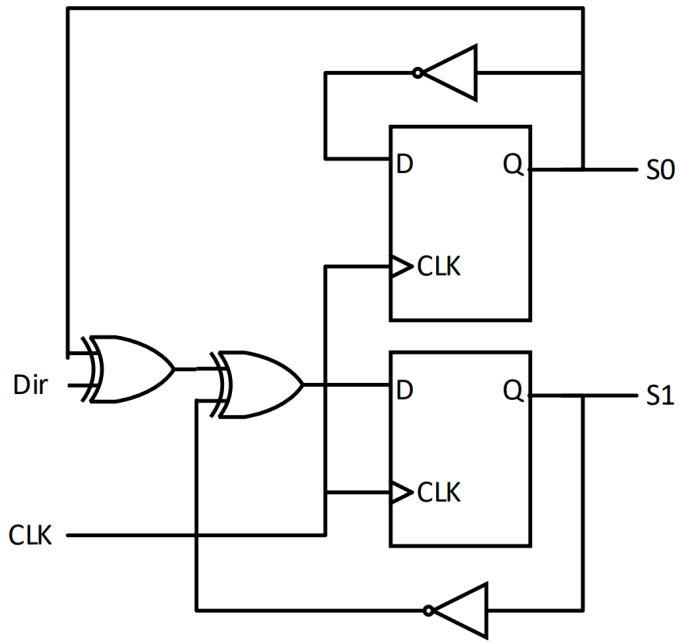




Exercício 1

- Tendo em conta o circuito apresentado:
  - Determine, a tabela de transição de estados e de saídas

$$D0 = \overline{Q0}$$
$$D1 = Dir \oplus Q0 \oplus \overline{Q1}$$
$$S0 = Q0$$
$$S1 = Q1$$



Exercício 1

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$$D0 = \overline{Q0}$$
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$$S0 = Q0$$
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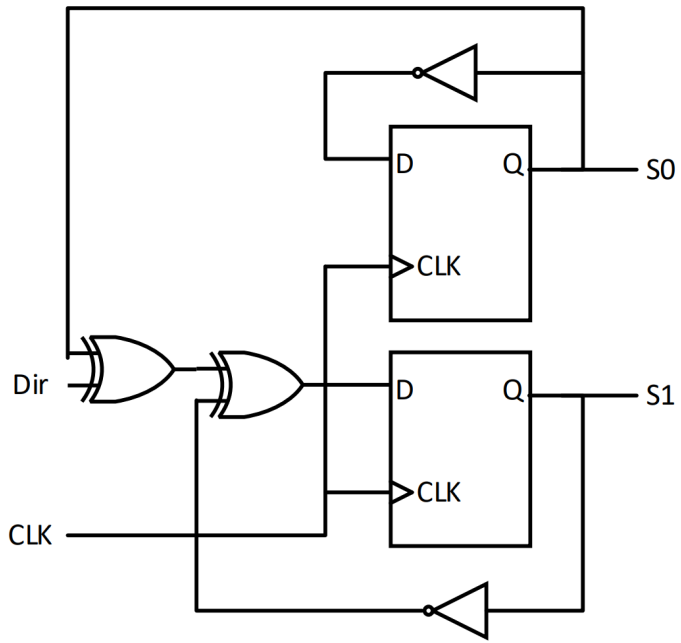
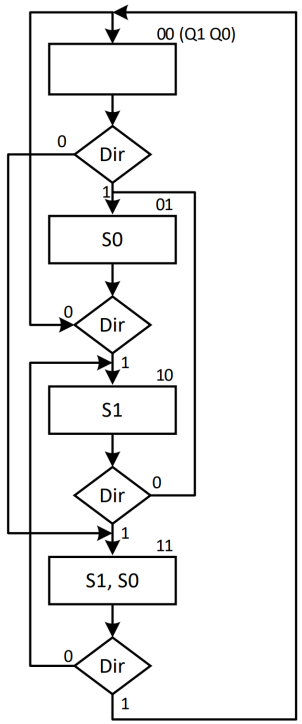
EP		Dir	ES		S1	S0
Q1	Q0		D1	D0		
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	1	0	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	1	0
1	0	1	1	1	1	0
1	1	0	1	0	1	1
1	1	1	0	0	1	1



Exercício 1

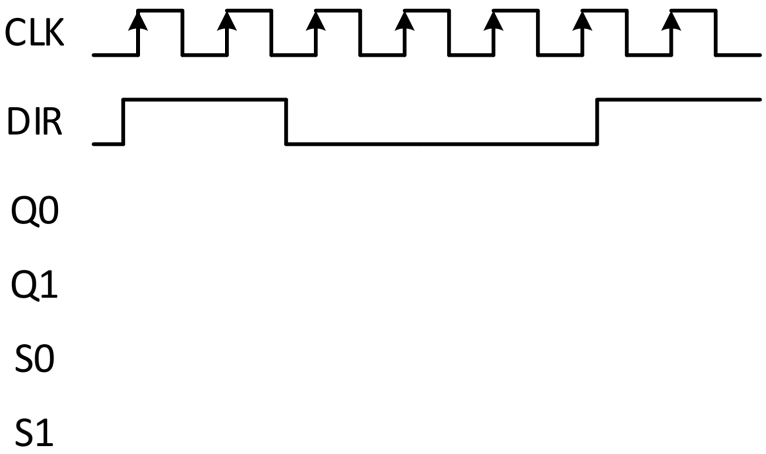
- Tendo em conta o circuito apresentado:
  - Determine, o ASM chart da máquina de estados

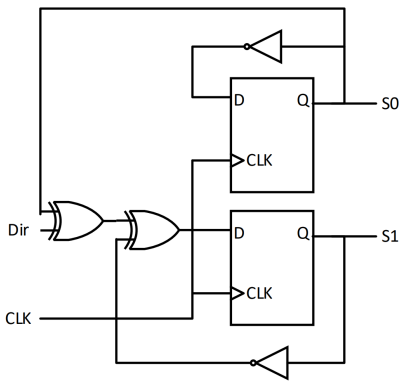
EP		Dir	ES		S1	S0
Q1	Q0		D1	D0		
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	1	0	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	1	0
1	0	1	1	1	1	0
1	1	0	1	0	1	1
1	1	1	0	0	1	1

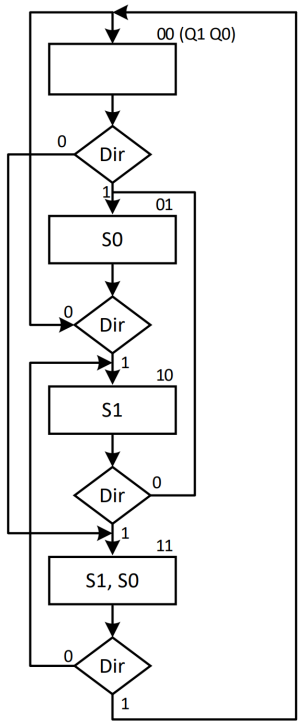


Exercício 1

- Tendo em conta o circuito apresentado:
  - Complete o diagrama temporal

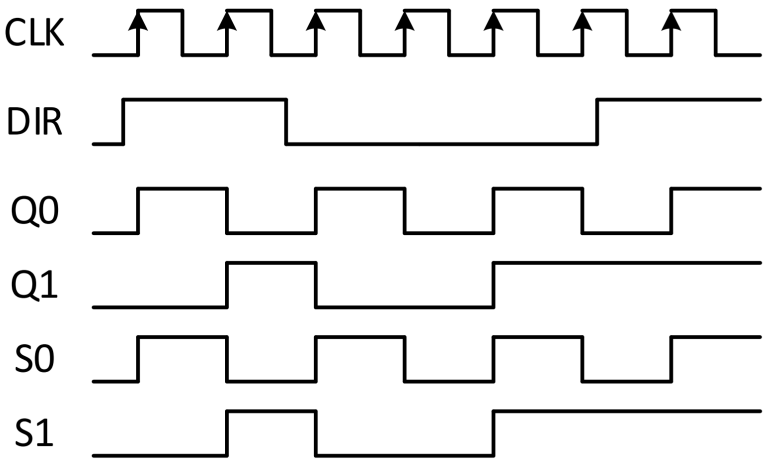


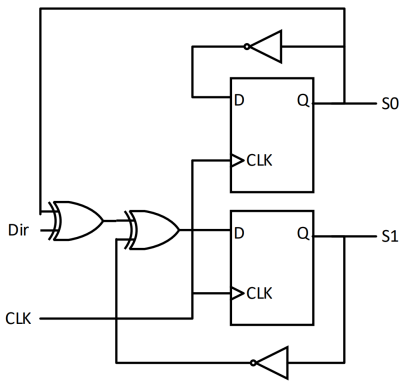


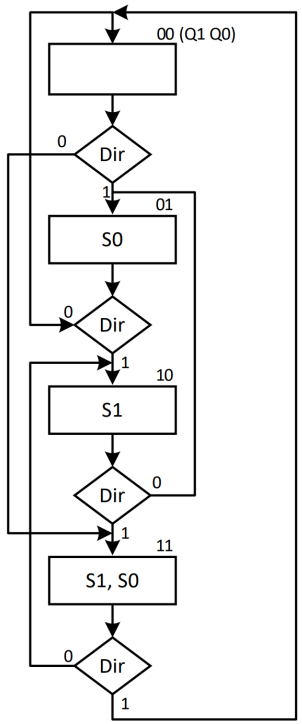


Exercício 1

- Tendo em conta o circuito apresentado:
  - Complete o diagrama temporal

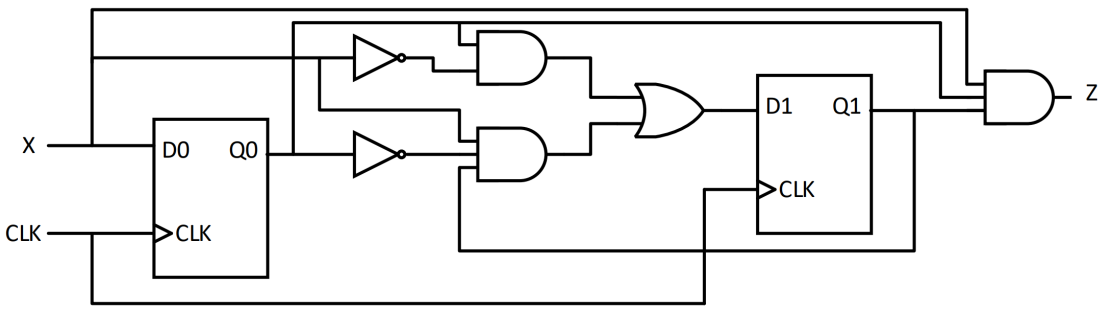


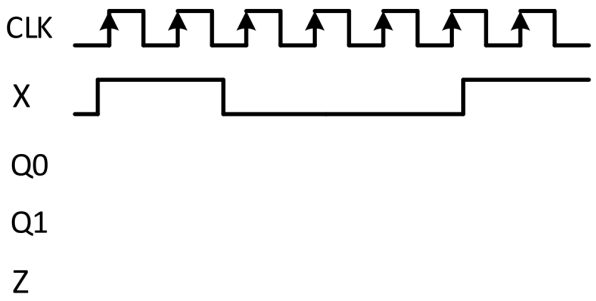




Exercício 2

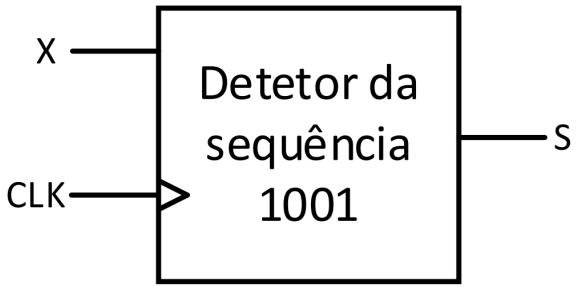
- Tendo em conta o circuito apresentado:
  - Determine, a tabela de transição de estados e de saídas
  - Determine, o ASM chart da máquina de estados
  - Complete o diagrama temporal





Exercício 3

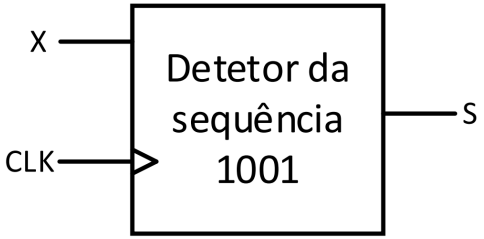
- Desenhe um circuito que identifique a ocorrência da sequência de valores binários 1001 na sua entrada;
  - Quando a sequência é detetada, a saída deve ficar ativa (com o valor lógico 1)
  - O circuito lê um bit da sua entrada a cada ciclo de relógio

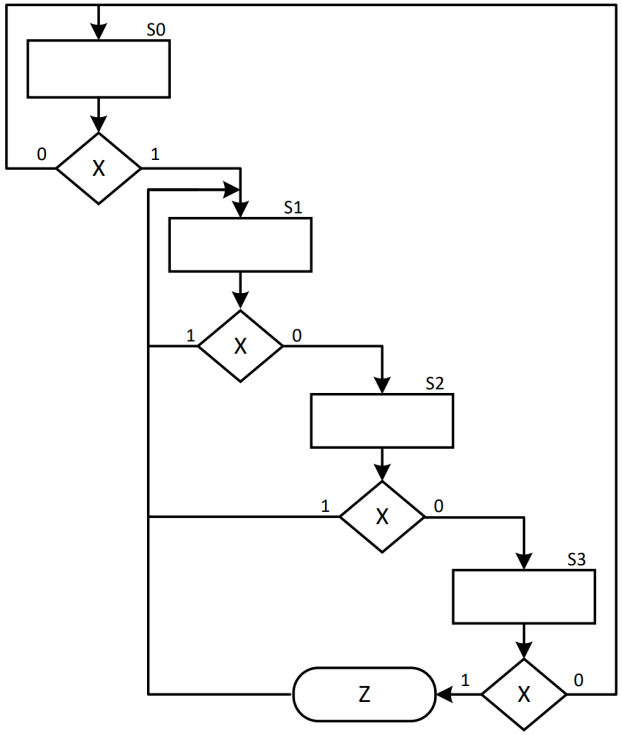


  - Exemplo de sequência:
    - X - 00100100100101100010010010010011110 1
    - S - 000001001001000000000100100100100000

Exercício 3

- Desenhe um circuito que identifique a ocorrência da sequência de valores binários 1001 na sua entrada;
  - Exemplo de sequência:
    - X - 00<u>1</u>00<u>1</u>00<u>1</u>001011000<u>1</u>00<u>1</u>00<u>1</u>00<u>1</u>00111101
    - S - 00000100100100000000010010010010000 0

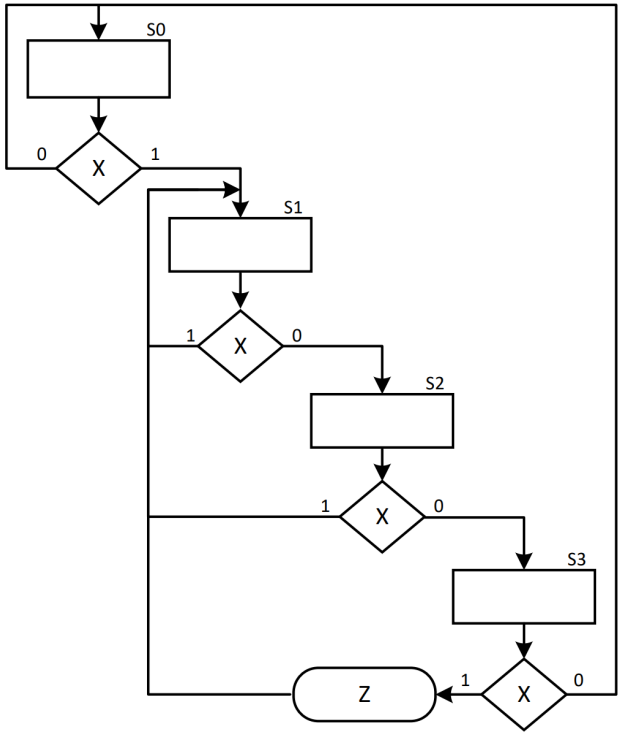




Exercício 3

- Obter a tabela de transição de estados e saídas

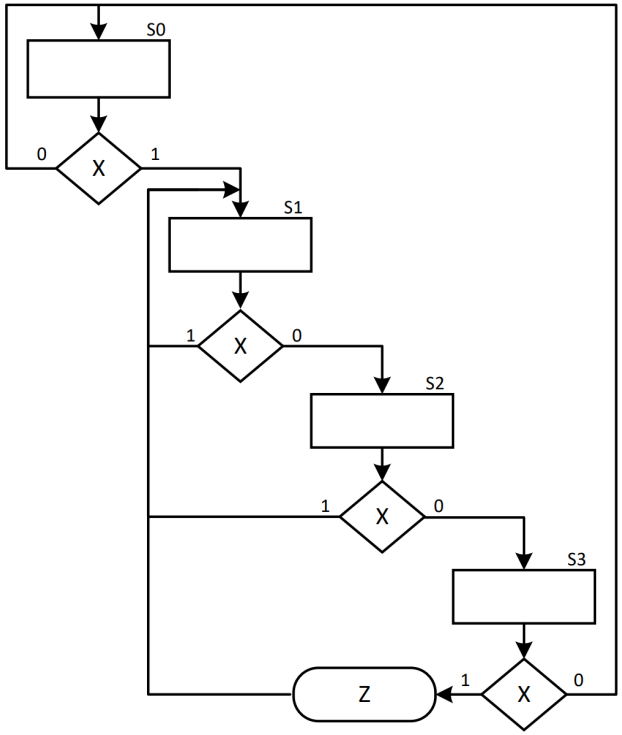
EP	X	ES	Z
S0	0		
S0	1		
S1	0		
S1	1		
S2	0		
S2	1		
S3	0		
S3	1		



Exercício 3

- Obter a tabela de transição de estados e saídas

EP	X	ES	Z
S0	0	S0	0
S0	1	S1	0
S1	0	S2	0
S1	1	S1	0
S2	0	S3	0
S2	1	S1	0
S3	0	S0	0
S3	1	S1	1

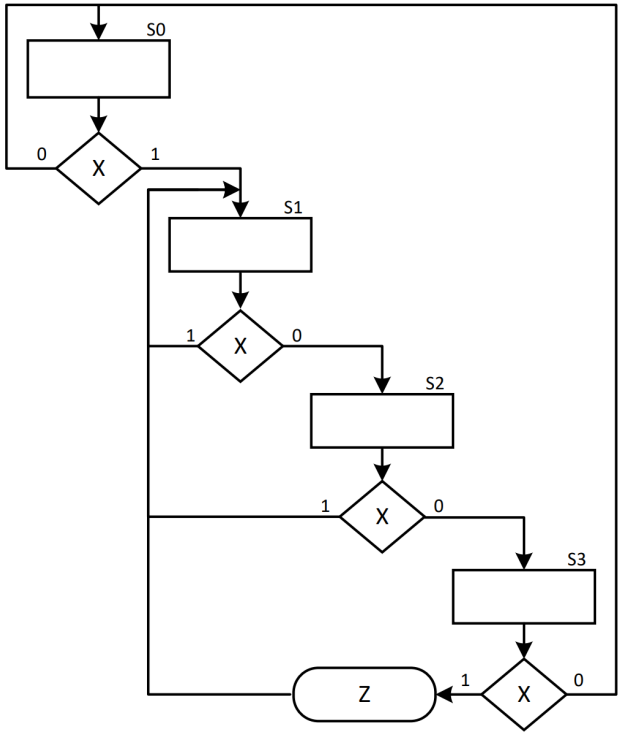


Exercício 3

- Obter a tabela de transição de estados e saídas
  - Codificar os estados

Estado (nome)	Codificação	
	Q1	Q0
S0	0	0
S1	0	1
S2	1	0
S3	1	1

EP		X	ES		Z
Q1	Q0		D1	D0	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1



Exercício 3

- Obter a tabela de transição de estados e saídas
  - Obter expressões

EP		X	ES		Z
Q1	Q0		D1	D0	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1

Q1 \ Q0	0	0	0	1
1	1	0	0	0

X

$$D1 = Q1\,\overline{Q0}\,\overline{X} + \overline{Q1}\,Q0\,\overline{X}$$

Q1 \ Q0	0	1	1	0
1	1	1	1	0

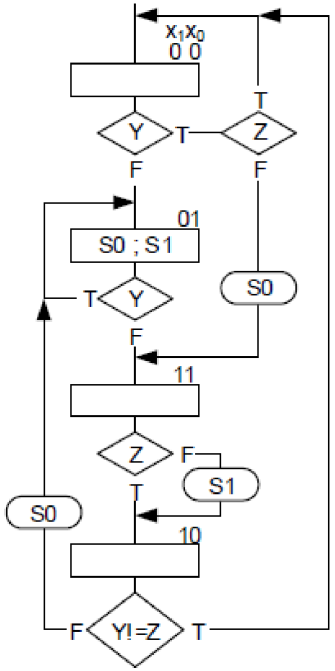
X

$$D0 = X + Q1\,\overline{Q0}$$

$$Z = Q1\,Q0\,X$$

Exercício 4

- Implemente a máquina de estados representada pelo seguinte ASM chart



Exercício 4

- Implemente a máquina de estados representada pelo seguinte ASM chart

EP X1 X0	Y	Z	ES D1 D0	S0	S1
0 0	0	0	0 1	0	0
0 0	0	1	0 1	0	0
0 0	1	0	1 1	1	0
0 0	1	1	0 0	0	0
0 1	0	0	1 1	1	1
0 1	0	1	1 1	1	1
0 1	1	0	0 1	1	1
0 1	1	1	0 1	1	1
1 0	0	0	0 1	1	0
1 0	0	1	0 0	0	0
1 0	1	0	0 0	0	0
1 0	1	1	0 1	1	0
1 1	0	0	1 0	0	1
1 1	0	1	1 0	0	0
1 1	1	0	1 0	0	1
1 1	1	1	1 0	0	0

D1

X1 \ X0	Y=0, Z=0	Y=0, Z=1	Y=1, Z=1	Y=1, Z=0
X0=0	0	0	0	1
X0=1	1	1	0	0
X1=1	1	1	1	1
X1=0	0	0	0	0

D0

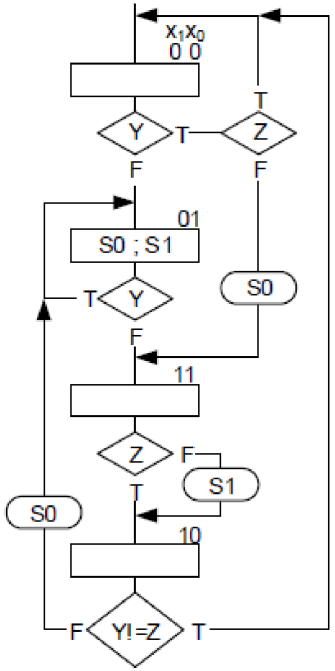
X1 \ X0	Y=0, Z=0	Y=0, Z=1	Y=1, Z=1	Y=1, Z=0
X0=0	1	1	0	1
X0=1	1	1	1	1
X1=1	1	0	1	0
X1=0	0	0	0	0

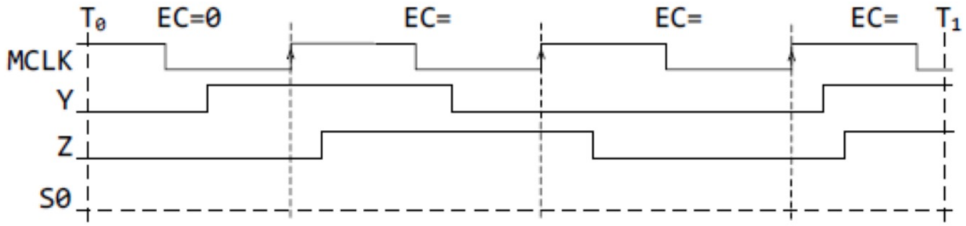
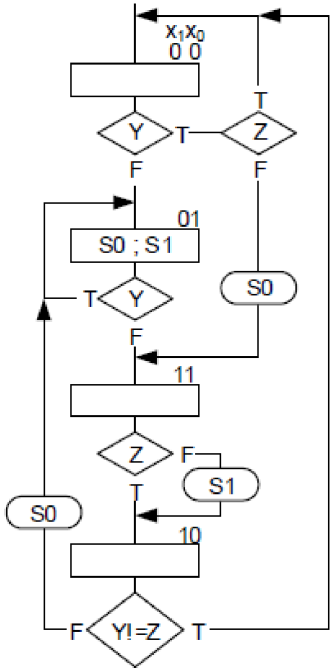
S0

X1 \ X0	Y=0, Z=0	Y=0, Z=1	Y=1, Z=1	Y=1, Z=0
X0=0	0	0	0	1
X0=1	1	1	1	1
X1=1	0	0	0	0
X1=0	1	0	1	0

S1

X1 \ X0	Y=0, Z=0	Y=0, Z=1	Y=1, Z=1	Y=1, Z=0
X0=0	0	0	0	0
X0=1	1	1	1	1
X1=1	1	0	0	1
X1=0	0	0	0	0



Exercício 4

- Análise temporal